Comparative Analysis of Data Augmentation Techniques for Enhancing Generalizability in Brain Tumor Classification with Limited Datasets

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***Abstract*—Deep learning models for brain tumor classification often face challenges related to generalizability, primarily due to the limited availability of labeled medical data. While data augmentation techniques offer a promising solution by expanding datasets artificially and potentially improving model performance, existing research lacks a comprehensive comparative analysis of their effectiveness in enhancing model generalizability. Furthermore, the literature lacks extensive discussions on the performance of models trained on limited datasets and their ability to generalize to unseen, potentially larger datasets. To address these gaps, this study investigates the generalizability of various data augmentation techniques for brain tumor classification with limited data. Specifically, we focus on two key problems: (1) the comparative analysis of model generalizability to unseen datasets, and (2) the comparison of augmentation techniques and their impact on model generalizability. We train a ResNet50 model on a small dataset and evaluate its performance on multiple datasets, augmented using techniques including geometric transformations, SMOTE, ADASYN, and GAN-based augmentation. Through an analysis of classification performance metrics such as accuracy, precision, recall, and F1-score, we aim to identify the most effective techniques for enhancing the generalizability of models trained on limited datasets.**

***Keywords*—Brain Tumor; Classification; MRI; Augmentation; Generalizability**

I. Introduction

The application of technology in the medical field is paramount, especially in situations where time is of the essence, and ensuring high accuracy is a matter of life and death. Rapid and precise diagnoses are crucial for determining the most effective treatment strategies and maximizing patient outcomes. However, the integration of machine learning into the medical domain faces significant challenges due to limited access to labeled medical data and stringent privacy concerns [2]. Ethical considerations surrounding patient data privacy and the potential for misuse necessitate careful methods for data collection and utilization.

One such challenge arises in the domain of medical image classification, specifically in the realm of brain tumor detection using MRI (Magnetic Resonance Imaging) scans. The accurate identification and classification of brain tumors based on image data plays a pivotal role in early diagnosis and the development of targeted treatment plans [10]. Early detection of brain tumors can significantly improve patient prognosis and survival rates. However, the development of robust machine-learning models for this task is hampered by the scarcity of annotated brain MRI images. The process of data annotation, which involves labeling images with the presence or absence of tumors, can be a time-consuming and expensive undertaking, requiring the expertise of trained medical professionals. Furthermore, privacy concerns surrounding patient data often limit the amount of data available for training and development purposes.

This research seeks to address the aforementioned challenge by harnessing the power of different data augmentation techniques. The motivation behind this research is deeply rooted in the critical need to enhance the accuracy and efficiency of brain tumor classification in MRI images, particularly within the context of medical imaging. The scarcity of annotated medical images, coupled with privacy concerns and ethical considerations, presents significant roadblocks in the development of robust machine-learning models for this task. Recent advancements in deep learning, notably the success of convolutional neural networks (CNNs), offer a promising avenue for addressing these limitations. By leveraging several techniques for data augmentation, this research seeks to create synthetic MRI images that closely resemble real patient data while preserving privacy. The ultimate goal is to study the model's generalizability to unseen datasets for different augmentation techniques. In essence, this research aims to investigate whether data augmentation can effectively mitigate the challenge of limited data and enhance the generalizability of brain tumor classification models, ultimately leading to more accurate diagnoses and improved patient outcomes.

A. Statement of Significance

Deep learning models for brain tumor classification struggle with limited labeled medical data, hindering generalizability. Data augmentation techniques can improve model performance, but a comprehensive comparison of their impact on generalizability for limited data is lacking. We investigate the generalizability of various data augmentation techniques for brain tumor classification with limited data. We focus on:

- **Comparative Analysis:** A comprehensive comparison

of how augmentation techniques affect model generalizability on unseen datasets.

- **Limited Data Performance:** Evaluating model performance on unseen datasets when trained with limited, augmented data

II. Literature Review

Recent studies have proposed applying data augmentation techniques to improve classification model performance. However, many of these studies fail to observe, discuss, and compare the effects of different augmentation methods specifically on brain tumor classification. Moreover, comparative analysis across different datasets is missing in most of the research which is crucial for the model's generalizability.

For instance, in a study [5], only two geometric augmentations (horizontal and vertical flips) were used. The authors employed a publicly available Contrast-Enhanced Magnetic Resonance Imaging (CE-MRI) dataset [8], containing four classes: glioma, meningioma, pituitary tumor, and healthy controls. Their evaluation of various models revealed a decrease in accuracy after applying these augmentation techniques, but they noted that pre-trained models enhanced results compared to CNN alone. Some studies [14, 15] incorporated multiple geometric transformations, including random rotations, translations, flipping, shearing, and zooming. Augmentation notably increased the accuracy of AlexNet, ResNet 50, and Inception V3 models, with AlexNet showing superior performance among the three.

In recent years, generative models such as GANs [11] have gained traction for synthetic data generation to augment datasets. In a study [4], a Generative Adversarial Network (GAN) approach was proposed to generate realistic brain tumor images, addressing limitations in current methods. Their method achieved an average accuracy of 96% across all four classes in brain tumor classification. Another approach, BrainGAN [3], utilized Vanilla GAN and DCGAN architectures to generate synthetic brain MRI images and augment the training dataset [7]. Results showed that training a ResNet152V2 model on DCGAN-generated images yielded the highest accuracy (99.09%), indicating DCGAN's ability to produce more realistic synthetic MRI scans compared to Vanilla GAN. In a deep learning framework [1] combining Variational Autoencoders (VAEs) and GANs, substantial improvements in brain tumor classification accuracy were reported, with the augmented dataset and ResNet50 model achieving 96.25% accuracy compared to 72.63% using only real data. Additionally, in another study [17], various GAN architectures (DCGAN, Cycle-GAN, Info-GAN, Conditional GAN, and Progressive Growing GAN) were evaluated, with Progressive Growing GAN (PPGAN) achieving the best results, including an accuracy of 98.8%, precision of 98.45%, and recall of 97.2%.

A recent study [13] investigated the impact of different loss functions, including focal loss and cross-entropy loss, and oversampling methods, such as Geometric, SMOTE, and ADASYN, in addressing the data imbalance issue for brain tumor classification tasks. Results indicated that VGG-16 consistently outperformed other models across all techniques in terms of accuracy evaluation. Geometric augmentation yielded the highest overall accuracy of 0.96, while SMOTE and ADASYN slightly improved the accuracy.

III. Methodology and Experiments

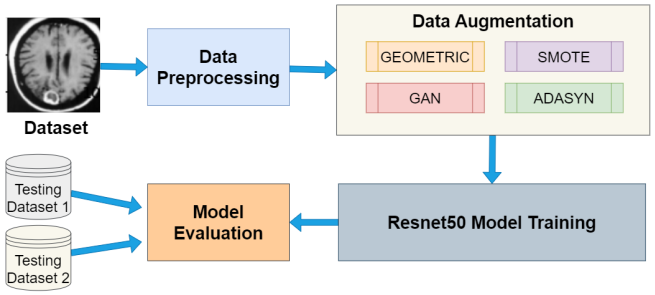


Fig. 1. Workflow Diagram

A. Data Collection and Preprocessing

We have used three publicly available datasets in our research and they will be referred to as Dataset1 [7], Dataset2 [8], and Dataset3 [6] in this article. Dataset 1 was used for training purposes and the rest were used to see the model's generalizability. The training Dataset used in this research comprises a total of 320 MRI images in the training set, categorized into two distinct classes: Normal and Tumor. Within the dataset, there are 136 MRI images in the Normal class, which serve as a reference for healthy brain scans, while the Tumor class contains 184 MRI images. Dataset2 is basically a multiclass dataset with 3 different tumor types and also contains normal images. We have merged all tumor types into one category i.e. Tumor. Dataset3 is mainly for the localization of brain tumors but we have used it for the classification task as Tumor and Non-Tumor classes are labeled. After acquiring the training data we normalized it and further resized it into 256x256 dimension. We employed ResNet50 architecture which requires the images to be a 3-channel input. To accommodate this, the single-channel images are preprocessed by replicating the channel information to create RGB images.

TABLE I. Distribution of Tumor and Non-Tumor Samples Across Datasets

Dataset	Tumor	Non-Tumor	Total
Dataset1 (Training)	184	136	320
Dataset2 (Testing)	289	105	394
Dataset3 (Testing)	1500	1500	3000

B. Model Architecture

Using pre-trained architectures can enhance results compared to using CNN alone [5]. We also tested it on our dataset and found the pre-trained model worked significantly well with respect to training a CNN model from scratch. ResNet50 is a powerful convolutional neural network (CNN) that has been extensively trained on the vast ImageNet dataset for image classification tasks. This pre-trained model serves as a foundation for the brain tumor classification task in our research. To adapt ResNet50 to the specific challenge

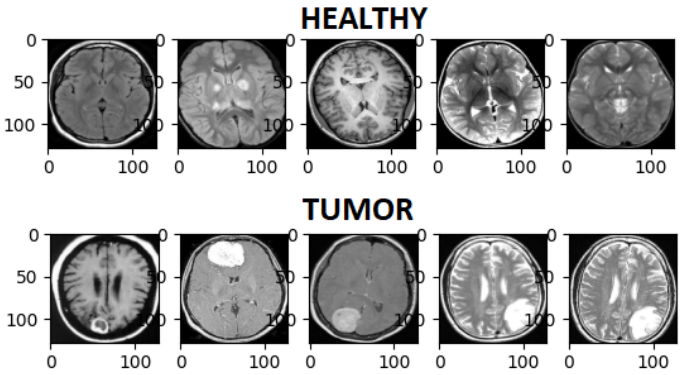


Fig. 2. Tumor and Healthy image samples from Dataset1

of brain tumor detection, the model is loaded without its final classification layers. Additionally, the pre-trained weights are initialized from a file, allowing the model to leverage the extensive image recognition knowledge learned from ImageNet. However, brain scans typically consist of a single channel, unlike the RGB format (red, green, blue) used in ImageNet. To address this difference, the single-channel brain image data undergoes preprocessing. This preprocessing step replicates the channel information, essentially converting the single-channel data into a format compatible with the pre-trained ResNet50 architecture. To tailor the model for the specific binary classification problem (tumor vs. no tumor), custom top layers are added on top of the frozen pre-trained layers. These custom layers typically include a global average pooling layer to summarize feature maps and dense layers with ReLU activation for non-linearity. Finally, the model utilizes a softmax activation function in the final layer. This function outputs class probabilities for the two classes (tumor and no tumor), enabling the model to predict the likelihood of a brain tumor being present in the analyzed scan.

C. Augmentation Techniques

In this section, we will discuss various augmentation techniques used to train the model. We employed 4 distinct augmentation techniques i.e. geometric, SMOTE, ADASYN, and GAN-based augmentation.

1) Geometric: To enrich our training dataset and improve model robustness, we employed various data augmentation techniques using the ImageDataGenerator class. Images were randomly rotated (up to 15 degrees) to introduce variations in object orientation. Zooming (up to 10% range) simulated slight changes in image scale, and horizontal/vertical flips created variations in object position within the frame. Additionally, spatial shifting (up to 10% of image dimensions) introduced minor spatial translations. Beyond geometric manipulations, we also explored color space augmentation. Brightness levels were randomly adjusted within a range of 0.8 to 1.2 times the original value to simulate lighting variations.

2) SMOTE: The second augmentation technique we employed is SMOTE (Synthetic Minority Oversampling Technique) [9]. SMOTE tackles the class imbalance issues issue by creating synthetic samples for the minority class. These synthetic data points are generated by interpolating between existing minority class samples, effectively increasing the representation of the minority class and mitigating the negative effects of class imbalance on model performance.

3) ADASYN: We also considered ADASYN (Adaptive Synthetic Sampling Approach) for data augmentation [12]. Unlike SMOTE's uniform oversampling, ADASYN focuses on generating synthetic data points for minority class samples that are harder for the model to learn. This targeted approach aims to improve model performance specifically for those challenging minority class instances.

4) DCGAN: GANs involve training two neural networks: a generator that creates new, realistic-looking data points, and a discriminator that tries to distinguish real data from the generated data. We used DCGAN architecture for this purpose and generated synthetic images from the model [16]. The discriminator is trained independently initially. It is provided with real MRI images and images generated by the current state of the generator. The binary cross-entropy loss is used to optimize the discriminator's ability to correctly classify images as real or generated. The GAN is trained by combining the generator and discriminator into a single model. During training, the discriminator is frozen, and the generator is trained to generate images that can fool the discriminator. The GAN is optimized using the binary cross-entropy loss. Finally, the GAN model is compiled with the Adam optimizer and binary cross-entropy loss. The discriminator's optimizer and loss function are separately defined. The training process involves the continuous refinement of the generator and discriminator through adversarial training. We trained this model for 10,000 epochs on Kaggle T4 GPU. We generated 260 synthetic images for tumor and non-tumor classes. We further merged this data with the original training set and used the resultant set for training.

D. Training Setup

Our model leverages transfer learning with a pre-trained ResNet50 architecture for brain tumor classification on single-channel images. The training process unfolds in several key stages. First, we capitalize on the power of a pre-trained ResNet50 model, loaded with weights learned from the vast ImageNet dataset. We exclude the final classification layers of ResNet50, retaining its extensive image recognition capabilities while adapting it to our binary classification task. The input shape is set to (256, 256, 3) to match the pre-trained model's expectation of RGB images.

TABLE II. MODEL PARAMETERS AND TRAINING CONFIGURATION

Parameter	Value
Input Shape	(256, 256, 3)
Base Model	Res-Net50
Loss Function	Categorical Cross-Entropy
Optimizer	Adam
Learning Rate	0.0001
Evaluation Metric	Accuracy
Batch Size	16
Max Epochs	100
Early Stopping	Enabled (monitors validation loss)

We then combine the pre-trained base model with our custom top layers to create a new model. This model is compiled with the categorical cross-entropy loss function, suitable for multi-class classification with one-hot encoded

labels. The Adam optimizer with a learning rate of 0.0001 is used to update the model's weights during training. Accuracy is chosen as the metric to track classification performance.

To prevent overfitting, we employ an EarlyStopping callback that monitors the validation loss. If the validation loss doesn't improve for a predefined number of epochs, the training process is halted. The model is then trained with preprocessed training images, one-hot encoded training labels, a batch size of 16, and a maximum of 100 epochs. Validation data consisting of preprocessed test images from Dataset1 and their corresponding labels are provided to assess performance on unseen data during training. Table II summarizes the model configurations.

The model was initially trained on Dataset1 without augmentation and then we evaluated its performance on Dataset2 and Dataset3. This was done to compare the results before and after applying augmentation. In the next step, we used geometric augmentations to train the model on Dataset1 and evaluated it on the rest. Similarly, the SMOTE, ADASYN, and DCGAN augmentations were used for training the model and then evaluated on the other datasets. Detailed results are discussed in the next section.

E. Evaluation Metrics

The selection of appropriate evaluation metrics is crucial for obtaining a comprehensive understanding of a machine learning model's performance. In our study, we employed a combination of metrics to assess our model's effectiveness: accuracy, precision, recall, and F1-score. Accuracy provides a general overview of how often the model makes correct predictions. However, it can be misleading in scenarios with imbalanced datasets. Precision and recall offer more granular insights. By considering these various metrics, we gained a deeper understanding of the model's strengths and weaknesses.

IV. Results and Discussion

This section presents the results obtained from evaluating the brain tumor detection model, which was trained on Dataset1 using various augmentation techniques and then assessed on separate datasets (Dataset2 and Dataset3). We analyze the impact of these augmentation techniques on the model's performance metrics, including accuracy, precision, recall, and F1-score, aiming to comprehend the model's effectiveness and identify potential areas for improvement. Initially evaluated on the test set of Dataset 1, the trained model demonstrated impressive accuracy and an F1 score of 100%. It was further evaluated on unseen and comparatively larger datasets to gauge the model's generalizability. Table 2 illustrates the performance of the brain tumor detection model on Dataset2 and Dataset3, showcasing the results for each dataset and augmentation type across four key performance metrics: Accuracy, Precision, Recall, and F1.

A. Model's Performance without Augmentation

When no augmentation was applied, the model achieved an accuracy of 84.77% on Dataset2 and 81.2% on Dataset3. Dataset2 exhibited better performance in the positive class, with a recall of 0.98, but struggled with predicting the negative class, resulting in a precision of 0.84, indicating the presence of false positives. In Dataset3, both accuracy and F1 score were lower due to a high number of false positives, resulting in a precision of 74%. However, the recall remained high at 0.97. Although the model without augmentation performed better on Dataset 1, significant results were not achieved.

TABLE III. Comparison of Tumor and Non-Tumor Sample Counts using Different Data Augmentation Techniques

Method	Tumor	Non-Tumor
Original Data	184	136
SMOTE	184	184
ADASYN	186	184
GAN	284	296

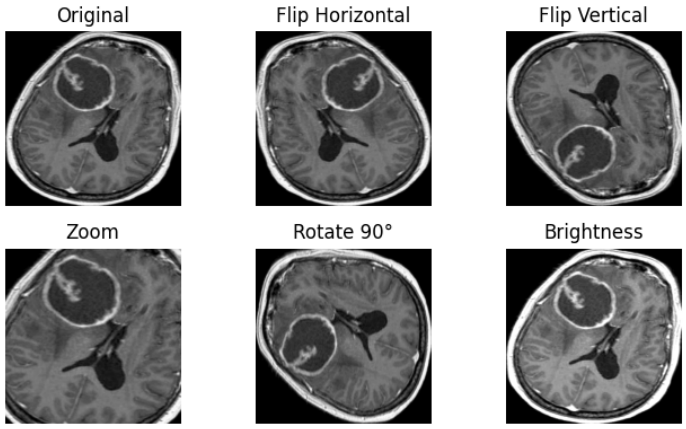


Fig. 3. Sample images generated using Geometric Augmentation

B. Model's Performance using Augmentation

Utilizing geometric augmentation techniques led to a slight improvement in model performance. The accuracy increased to 86.29% for Dataset2 and 83.97% for Dataset3. Precision remained relatively consistent for Dataset2 and increased by 4% for Dataset3, while recall maintained high values. Consequently, F1 scores rose to 0.91 for Dataset2 and 0.85 for Dataset3, indicating a balanced improvement in both precision and recall. Figure 3 illustrates the augmented images after geometric transformations such as flip, rotate, zoom, and brightness adjustments have been applied.

Applying SMOTE resulted in comparable performance to geometric augmentation. The accuracy for Dataset2 reached 84.26%, with precision, recall, and F1 scores closely aligned with those achieved without augmentation. Similarly, for Dataset3, the accuracy increased to 84%, with precision and recall maintaining similar levels to other augmentation techniques. Figure 4 illustrates the synthetic samples generated using ADASYN, showing how the algorithm creates additional minority class examples based on feature space density.

ADASYN augmentation exhibited similar trends to SMOTE. Although the accuracy for Dataset2 slightly increased to 84.5%, the precision, recall, and F1 scores remained consistent with other augmentation methods. However, for Dataset3, the accuracy, precision, and recall were slightly lower compared to the other augmentation techniques. Figure 4 demonstrates the synthetic samples generated using SMOTE,

TABLE IV. PERFORMANCE COMPARISON OF DIFFERENT AUGMENTATION TECHNIQUES ON TWO DATASETS

Augmentation Type	Dataset2				Dataset3			
	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1
No Augmentation	84.77	0.84	**0.98**	0.90	81.2	0.74	**0.97**	0.84
Geometric	**86.29**	**0.86**	0.97	**0.91**	83.97	**0.78**	0.95	0.85
SMOTE	84.26	**0.86**	0.94	0.90	**84**	**0.78**	0.96	**0.86**
ADASYN	84.5	0.84	0.97	0.90	82.8	0.76	0.96	0.85
DCGAN	80.71	0.81	0.95	0.88	73.5	0.66	**0.97**	0.78

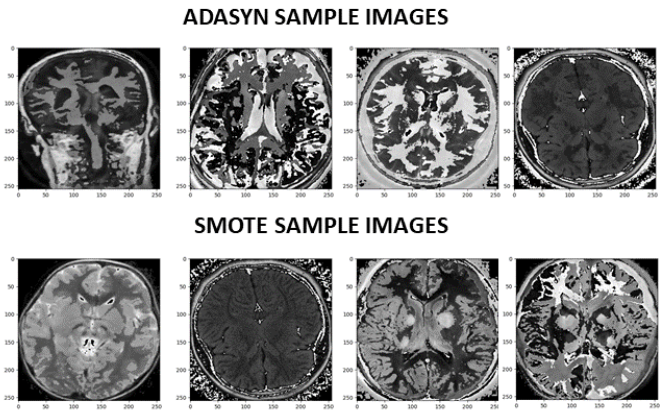


Fig. 4. Sample images generated using SMOTE and ADASYN

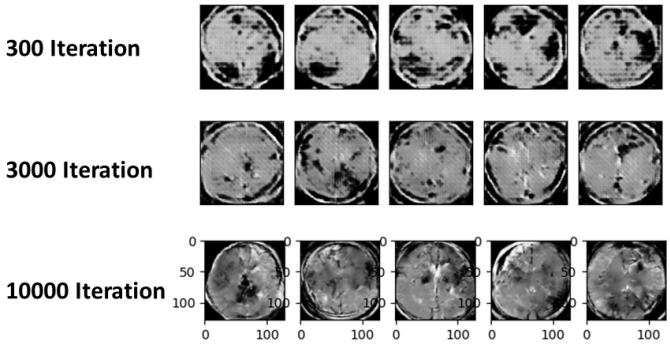


Fig. 5. GAN generated samples from some selected iterations

where new instances are interpolated between existing minority class samples to balance the dataset.

In contrast, employing DCGAN for augmentation led to a notable decrease in performance across both datasets. The accuracy dropped to 80.71% for Dataset2 and 73.5% for Dataset3. While recall remained relatively stable, precision experienced a considerable decrease, resulting in lower F1 scores compared to other augmentation techniques. Figure 5 displays the images generated by the GAN after 300, 1000, and 10,000 iterations, illustrating the progression and refinement of image quality over the training process.

C. Discussion

The results of our study demonstrate that geometric augmentation and SMOTE augmentation consistently improved model performance across both datasets, contributing to enhanced accuracy while maintaining a harmonized balance between precision and recall. ADASYN augmentation also showcased a slight performance improvement, albeit comparable to the other two techniques. Conversely, the utilization of GAN for augmentation resulted in decreased performance, even comparable to results obtained without augmentation. This decline can be attributed to the limited quality and diversity of synthetic samples generated by the GAN, which may introduce artifacts or unrealistic tumor structures, negatively impacting the model's ability to generalize. Additionally, GAN training is computationally expensive, requiring significant resources, fine-tuning, and large datasets to generate high-quality samples. Given the small dataset size in our study, the GAN struggled to produce meaningful variations, ultimately leading to suboptimal augmentation results. Throughout all experiments, a notable challenge observed was the misclassification of the Non-Tumor class, primarily due to false positives, while false negatives remained relatively low. This discrepancy could be attributed to a slight data imbalance within the training dataset, with the Tumor class comprising 57% and the Non-Tumor class 43%. Consequently, the original model exhibited a bias toward the Tumor class, affecting its ability to classify Non-Tumor instances accurately. Furthermore, the presence of T1 and T2-weighted MRI images necessitates a balanced representation within the training set to facilitate better generalization. Similarly, the incorporation of various MRI image orientations, such as sagittal, axial, and coronal, underscores the importance of an even distribution of image types for improved model generalizability. Ensuring a balanced representation of both image types and orientations within the training set is essential for enhancing the model's ability to accurately classify brain tumor instances across diverse datasets. In summary, addressing data imbalance and ensuring a diverse representation of MRI image types and orientations are crucial steps toward enhancing the generalizability and performance of brain tumor classification models trained on limited datasets.

V. CONCLUSION AND FUTURE WORK

Our study investigated the impact of various augmentation techniques on the performance of a brain tumor detection model trained on limited datasets. Our findings highlight the effectiveness of geometric augmentation and SMOTE augmentation in consistently improving model performance, as

evidenced by enhanced accuracy and a balanced trade-off between precision and recall. Conversely, employing GAN for augmentation led to a notable decrease in performance.

Challenges observed during our experiments included the misclassification of the Non-Tumor class, primarily due to false positives, highlighting the presence of data imbalance within the training dataset. Addressing this imbalance and ensuring a balanced representation of both tumor and non-tumor instances is crucial for mitigating bias and improving model robustness. Additionally, the diversity of MRI image types (T1 and T2 weighted) and orientations further underscores the importance of an even distribution of image representations within the training set to facilitate better generalization. Moving forward, future research efforts should focus on exploring additional strategies for mitigating bias, improving model generalizability, and enhancing performance in brain tumor classification tasks on limited datasets. Other classification models like ResNet152, EfficientNet, InceptionV3, MobileNet, DenseNet, XceptionNet, etc. can also be utilized to compare the results with ResNet50. Furthermore, future research should focus on the balanced distribution of type and orientation of the MRI images.

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